7 Use case: LRU Cache

This chapter covers

- Solving a problem: avoid computing things twice
- Introducing caching as a solution
- Describing different types of caches
- Designing an efficient solution for LRU cache
- Discussing MFU and other choices to handle priority
- Discussing caching strategies
- Reasoning about concurrency and synchronization
- Describing how caches can be applied in a sentiment analyzer’s pipeline

This chapter is going to be different with respect to what you have seen so far in this book. We are not going to introduce a new data structure, but instead we will use the ones described in chapters 2 to 5 to create a more complex data structure; by using lists, queues, hash tables etc… as